GBIF | Global Biodiversity Information Facility

GBIF - Système Mondial d'Information sur la Biodiversité : Update from the GBIF Secretariat

Joe Miller, Directeur du Secretariat

Mélianie Raymond, Head of Community and Capacity

United Nations Conference on Environment and Development, Rio de Janeiro, Brazil, 1992

https://www.un.org/en/conferences/environment/rio1992

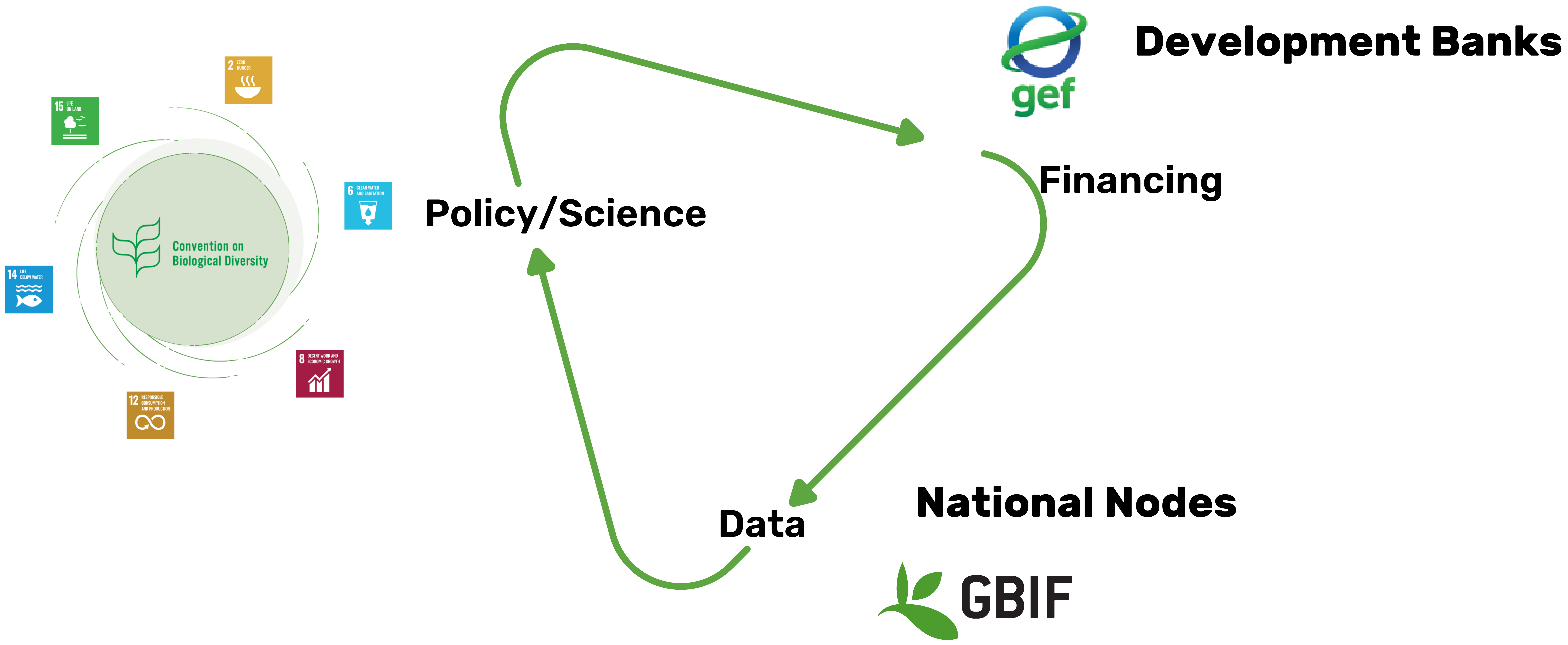
COP15 and the Global Biodiversity Framework and the GBF Fund

Convention on Biological Diversity

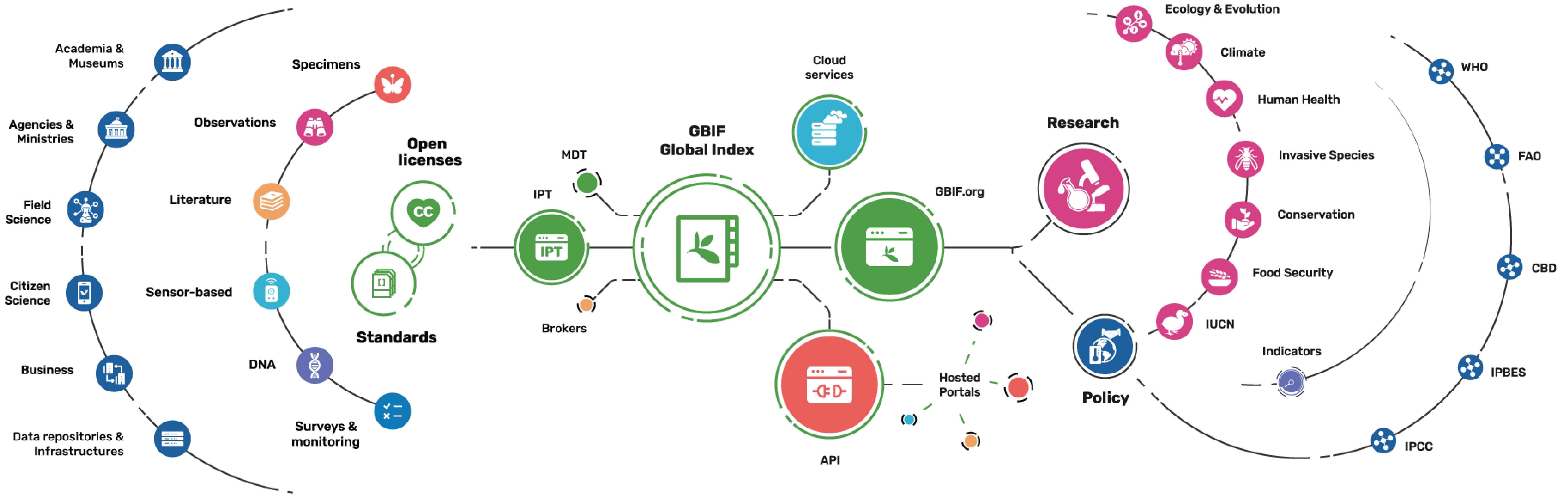
Convention on
Biological Diversity

Global Environment Facility

https://www.cbd.int/article/cop15-cbd-press-release-final-19dec2022


Convention on Biological Diversity
Policy/Science
Development Banks
gef
Financing
National Nodes
Data
GBIF

Providing biodiversity evidence for research and policy



Datasets

122,539

Country Participants

70

Countries sharing data

148

Publishers

2,717

Hosted portals

71

Peer-review papers using data

14,884

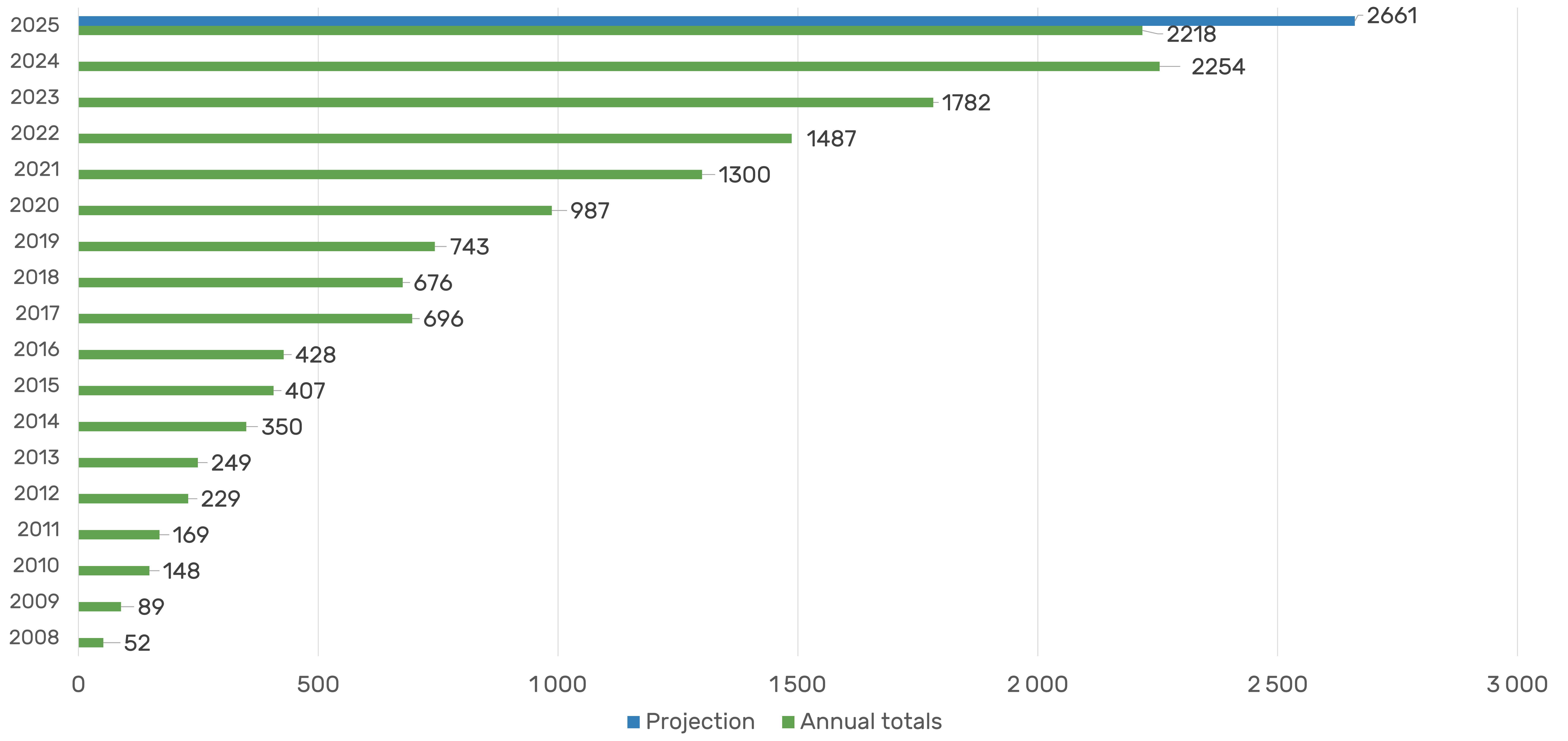
Average records downloaded per month (2025)

297 billion

Species occurrence records

3,737,061,311

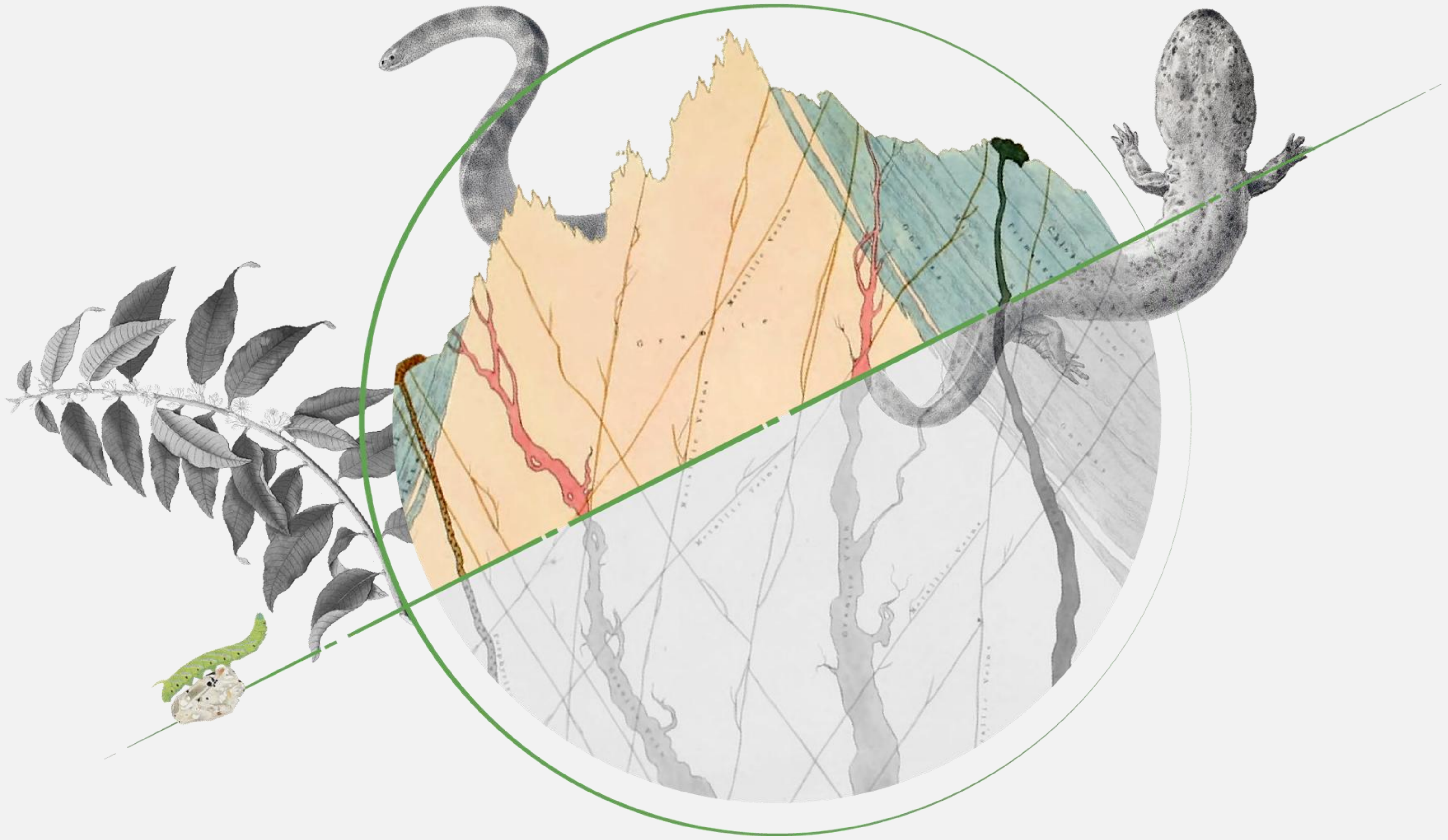
Peer-reviewed publications using GBIF-mediated data



https://www.gbif.org/resource/search?contentType=literature&literatureType=journal&relevance=GBIF_USED&peerReview=true

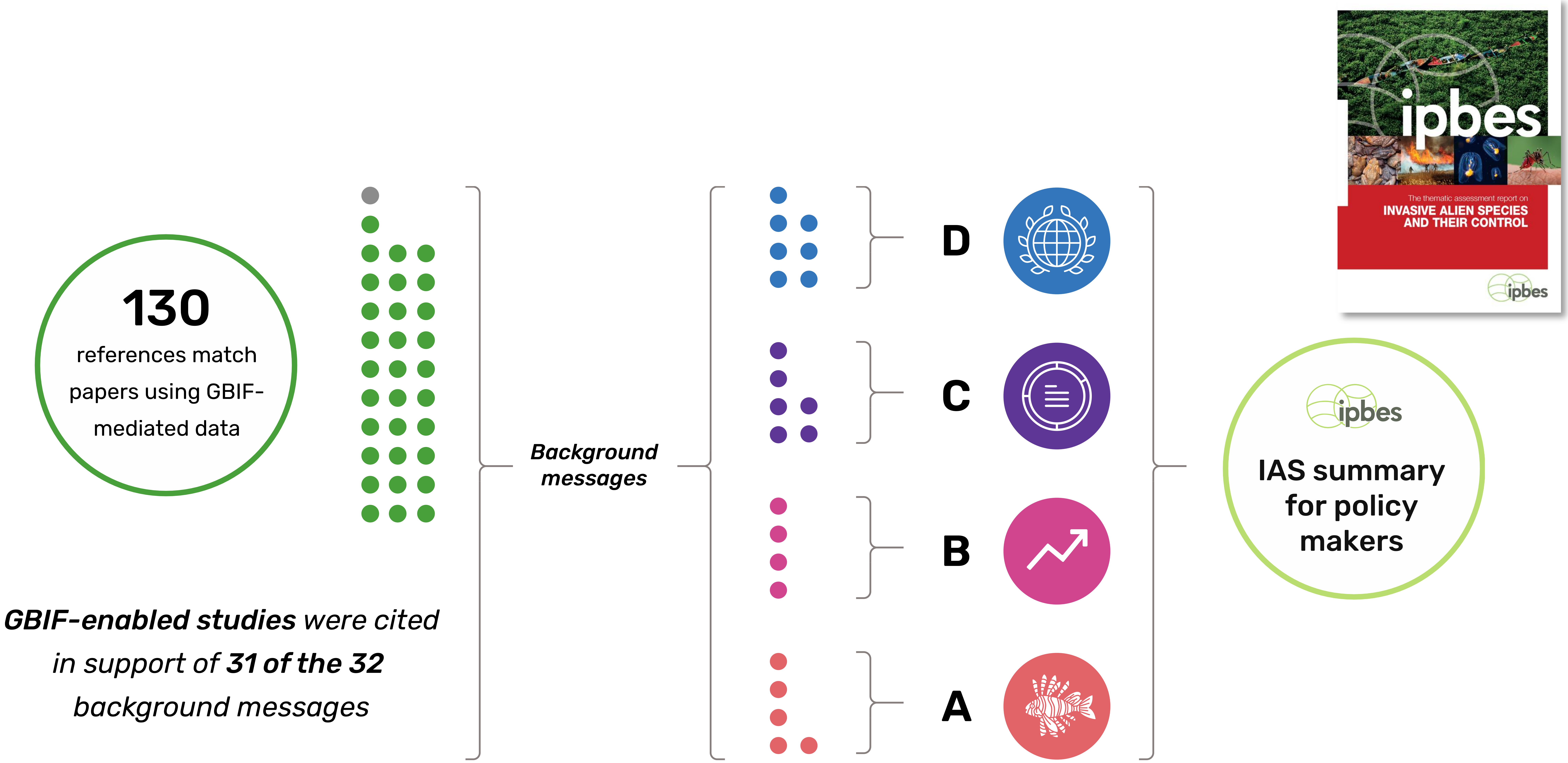
GBIF expands the scope of what is possible

Almost **half of GBIF users** report that they would have found it **impossible to achieve** the same outcomes in their work without GBIF



The economic value and impact of the GBIF network

For every **€1 invested** in GBIF, **users receive €3 of benefits** while **society gains up to €12**



doi.org/10.35035/3k5a-tv18

GBIF & the UN Convention on Biological Diversity

"Given the relevance of the work undertaken by GBIF to the implementation of the Kunming-Montreal Global Biodiversity Framework and to the Convention on Biological Diversity generally, I encourage all Parties to join GBIF as national participants"

Astrid Schomaker
Executive Secretary
—CBD Notification 2025-037

https://www.cbd.int/notifications/2025-037

GBIF mediated data supports CBD Global Biodiversity Framework

1 Data to identify key biodiversity areas

2 Data to monitor restoration

3 Data to locate & monitor protected areas

6 Data on invasive species occurrence

8 Data on climate change impact

4 Data for species conservation

15 Platform for sharing corporate biodiversity data

20 Strengthen capacity for data mobilization & use

21 Ensuring access to best-available data

https://www.cbd.int/gbf/

Data-driven biosecurity & alien invasive species reporting

- **Establish & implement guidelines** for managing data on alien invasive species
- Advance & **refine lists of regulated invasives** using IUCN GRIIS checklists
- Collaborate on developing data-driven **early-warning alert systems** & rapid-response mechanisms
- **Embed invasive risk analyses** into environmental impact assessments
- Incorporate biosecurity data into **NBSAPs & sectoral & cross-sectoral policies**

Component and complementary indicators

Global South collections help test new bioclimate model & invasive potential

- GBIF-enabled research tested a new bioclimate variable dataset by modelling global distribution of invasive plants
- New risk assessment resource relies on records from 100s of datasets in dozens of countries
- Data citation includes BID-mobilized records from Université de Lomé’s Collections des Herbiers d’Afrique de l'Ouest et d'Afrique Centrale

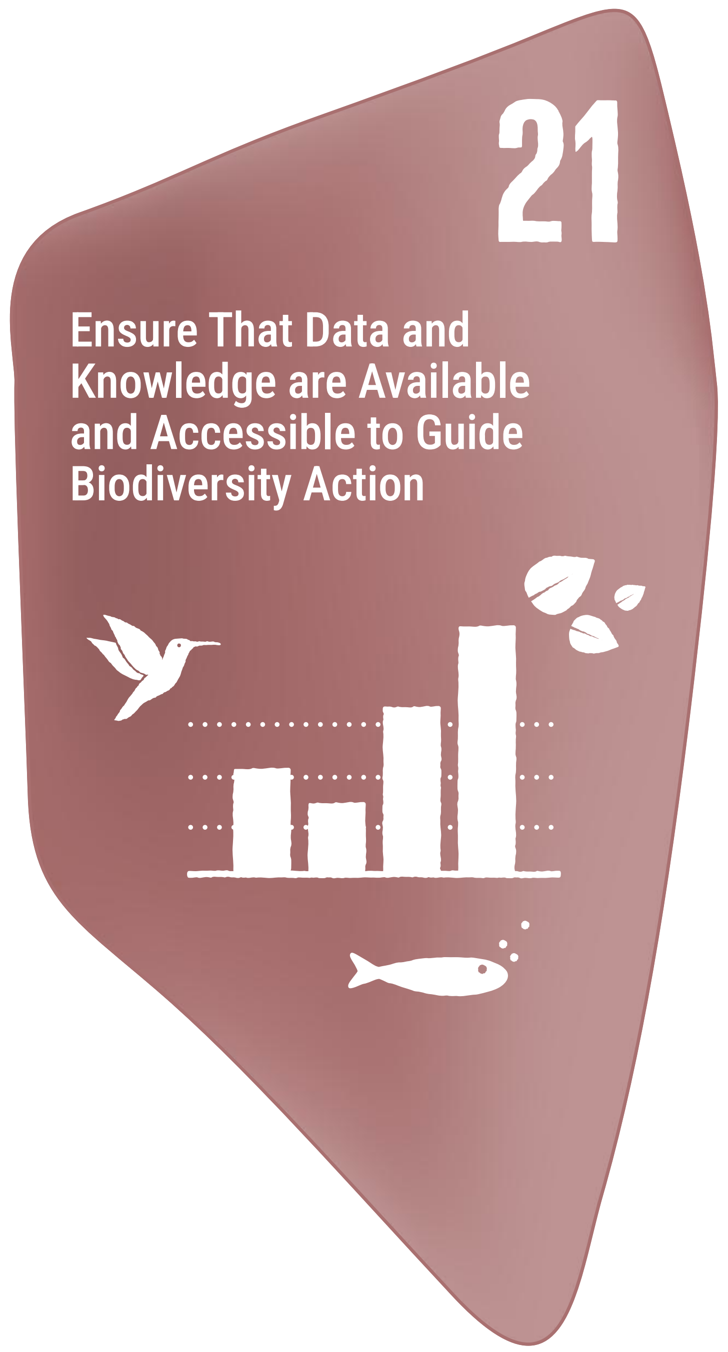
Zhang F, Wang C, Zhang C & Wan J (2023) Comparing the Performance of CMCC-BioClimInd and WorldClim Datasets in Predicting Global Invasive Plant Distributions. Biology 12(5): 652. https://doi.org/10.3390/biology12050652

African tulip tree (*Spathodea campanulata*), collected in Togo. https://www.gbif.org/occurrence/4066468495

Supporting national species data services

- Improve the accuracy and interoperability of species data
- With Catalogue of Life, support open-data infrastructure, services & training to:
- Build, standardize & manage lists of species names, including those used in national legislation
- Match occurrence data to the scientific names that countries use
- Harmonize synonymy and variation in names used at national & global scales
- Prepare & publish thematic lists of priority species for management, conservation and reporting

Component and complementary indicators



Increasing available data for IUCN Red List

- Assessment of Brazilian copal (*Hymenaea courbaril*) is one of 1,782 to cite use of GBIF-mediated data
- Report uses 9,317 records, including data from 46 BID-enabled datasets
- Sources come from 29 organizations from across Caribbean, Africa and Pacific regions

Bachman, S. 2023. *Hymenaea courbaril*. The IUCN Red List of Threatened Species 2023: e.T19891869A68101847. https://dx.doi.org/10.2305/IUCN.UK.2023-1.RLTS.T19891869A68101847.en.

Brazilian copal (*Hymenaea courbaril*), observed in French Guiana by Sébastien Sant, via iNaturalist. https://www.gbif.org/occurrence/4528063051 (CC BY-NC)

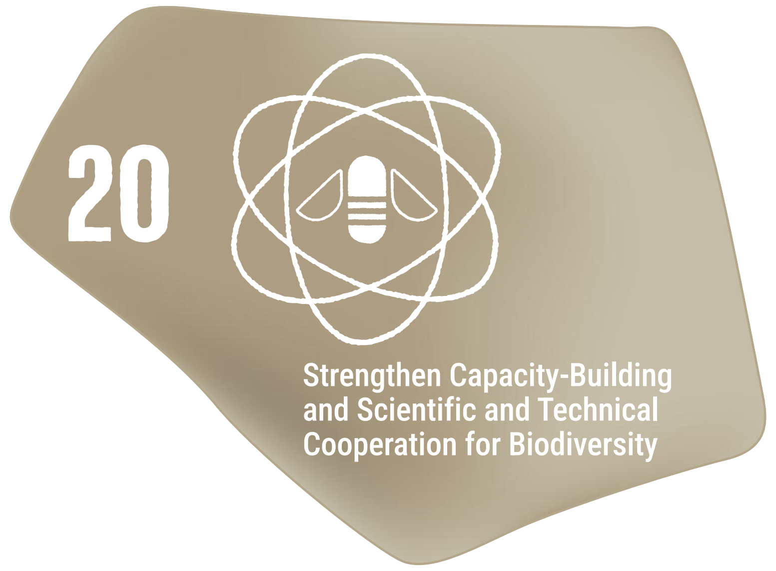
Integrated survey & monitoring data for reporting

- **Improve data quality & coverage** for target species monitored in ecosystems identified as priorities for restoration
- **Mobilize state-of-nature data** from LTER networks, field stations & restoration sites
- Provide dedicated training & support to **implement enhanced monitoring data standards**

Component indicators

French 7th National Reports to the CBD:

- GBIF occurrence data are linked to the capacity to measure 19 indicators
  - *Reducing threats to biodiversity*: Targets 2, 3, 5, 6, 7, 8 & 10
  - *Meeting people's needs through sustainable use and benefit-sharing:* Target 10
  - *Tools and solutions for implementation and mainstreaming:* Targets 15, 18 & 21
    - Biodiversity Credit schemes
    - Private Sector data sharing
- Opportunity for coordination for data needs for the 8th National report



GBIF as data infrastructure

GBIF as a community

Global nodes meeting 2025, Bogota Colombia

Community & Capacity team: No data without people!

Laura Russell, Maheva Bagard Laursen, Sanja Novakovikj

- Still important gaps in data availability
- Often these reflect gaps in participation

Capacity development objectives

Developing the capacity to mobilize biodiversity data through GBIF

The ability to capture, manage, and publish biodiversity data

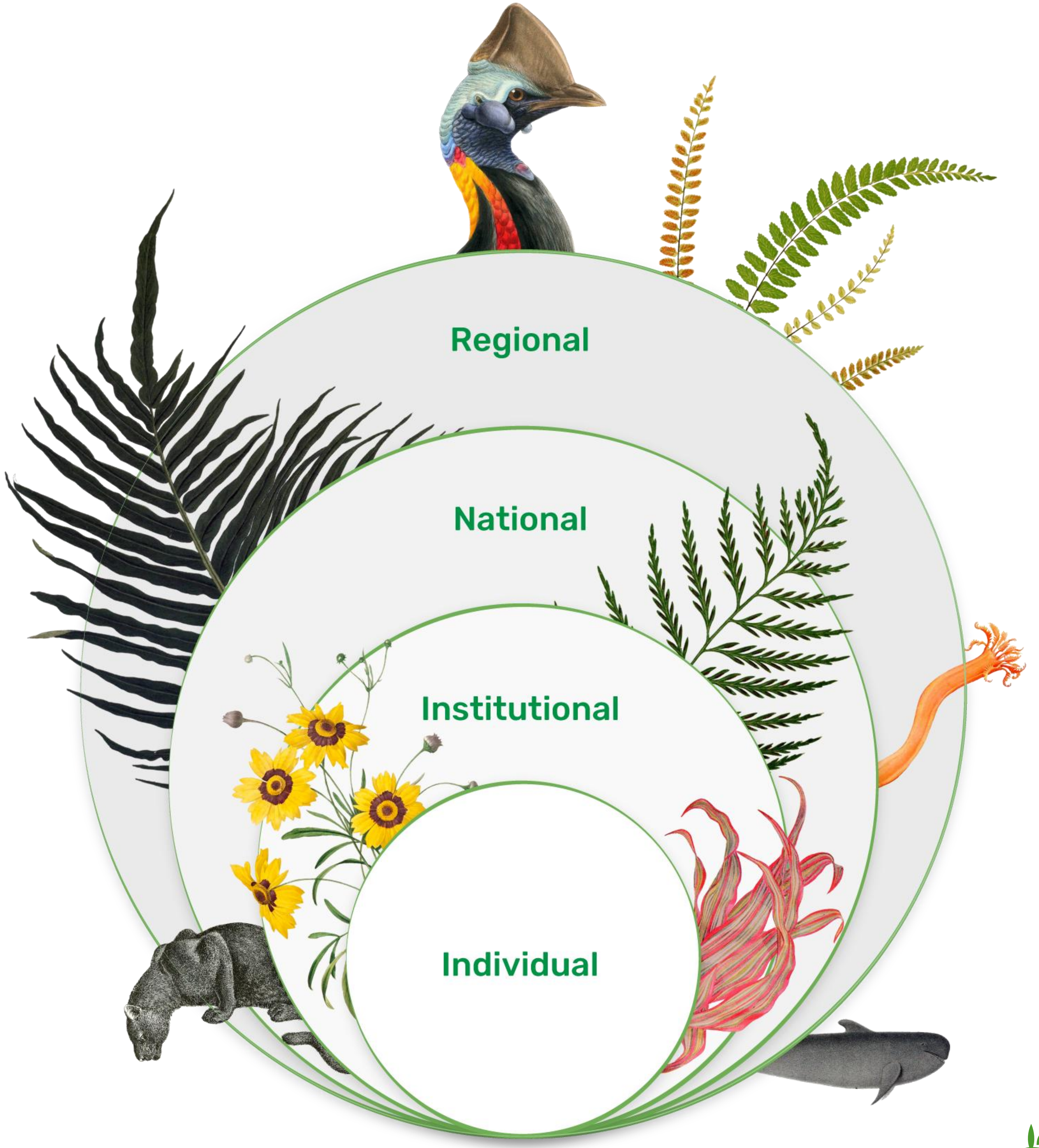
Developing the capacity to use GBIF-mediated data

The ability to analyze and use biodiversity data accessible through GBIF in scientific research and decision making

https://doi.org/10.35035/doc-wvg9-n274

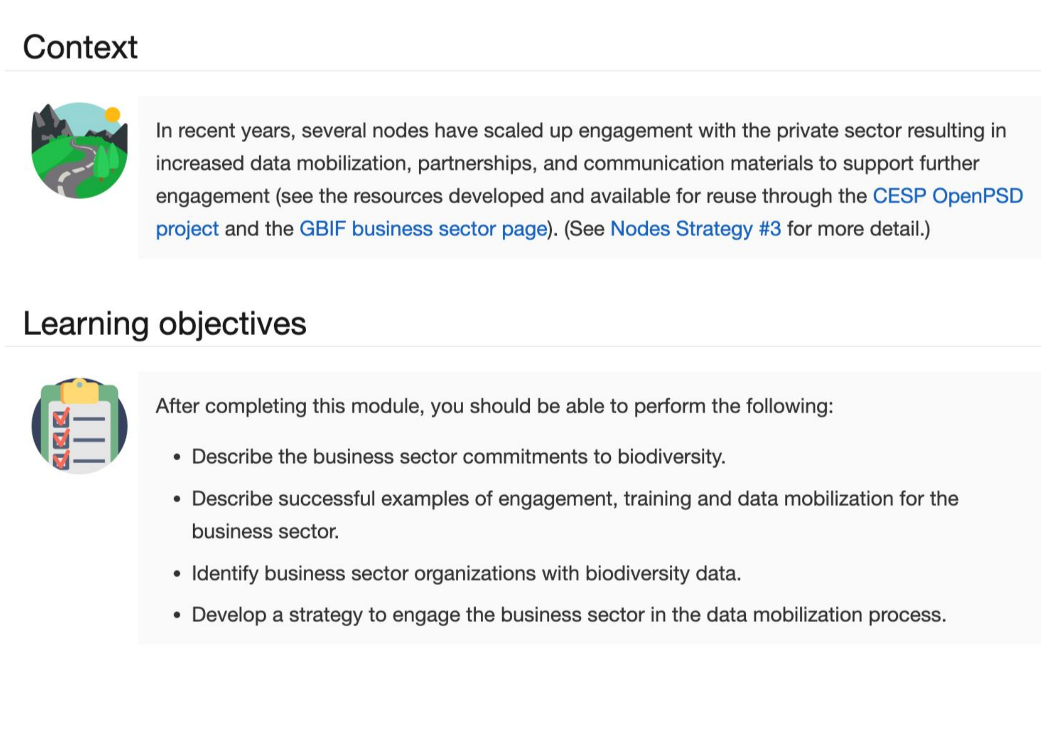
Scalable capacity development actions

Embedding capacity on nested levels in the network



Strengthening skills and opportunities for individuals to contribute to our work

Training strategy components


Context

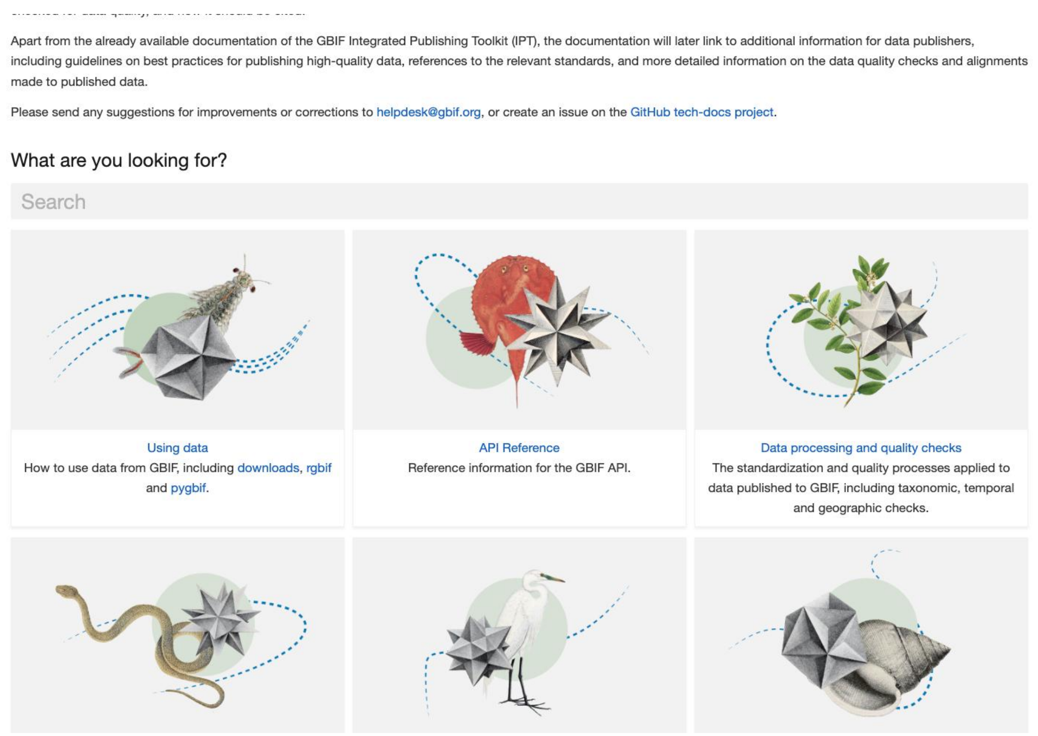
In recent years, several nodes have scaled up engagement with the private sector resulting in increased data mobilization, partnerships, and communication materials to support further engagement (see the resources developed and available for reuse through the CESP OpenPSD project and the GBIF business sector page). (See Nodes Strategy #3 for more detail.)

Learning objectives

After completing this module, you should be able to perform the following:

- Describe the business sector commitments to biodiversity.
- Describe successful examples of engagement, training and data mobilization for the business sector.
- Identify business sector organizations with biodiversity data.
- Develop a strategy to engage the business sector in the data mobilization process.

Curriculum development


Apart from the already available documentation of the GBIF Integrated Publishing Toolkit (IPT), the documentation will later link to additional information for data publishers, including guidelines on best practices for publishing high-quality data, references to the relevant standards, and more detailed information on the data quality checks and alignments made to published data.

Please send any suggestions for improvements or corrections to helpdesk@gbif.org, or create an issue on the GitHub tech-docs project.

What are you looking for?

Search

Using data
How to use data from GBIF, including downloads, rgbif and pygbif.

API Reference
Reference information for the GBIF API.

Data processing and quality checks
The standardization and quality processes applied to data published to GBIF, including taxonomic, temporal and geographic checks.

Digital documentation

Data use club

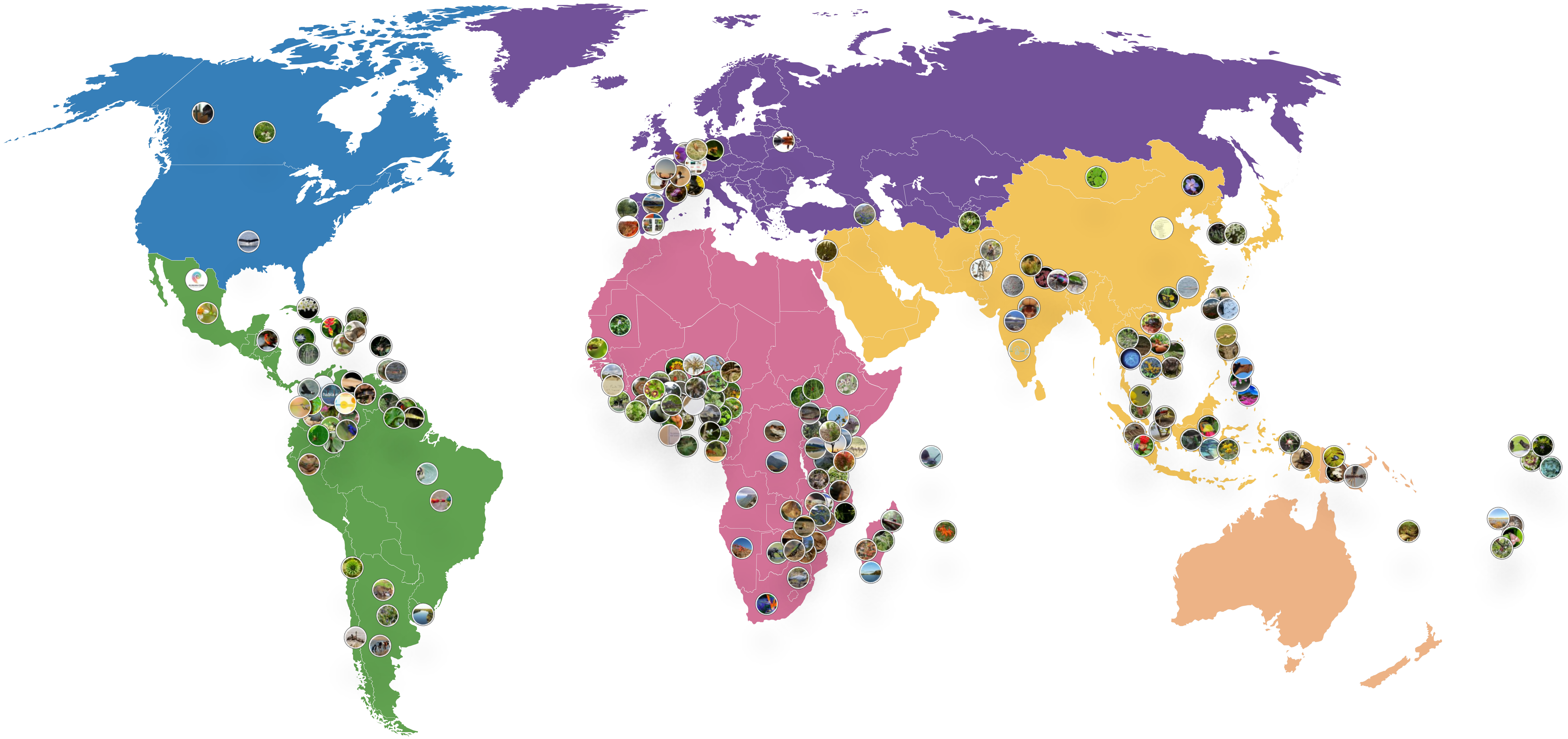
Community volunteers


BIBLIOTHEQUE
UNIVERSITAIRE
BIBLIO UNIVERSITAIRE


GBIF

Developing regional communities of practice

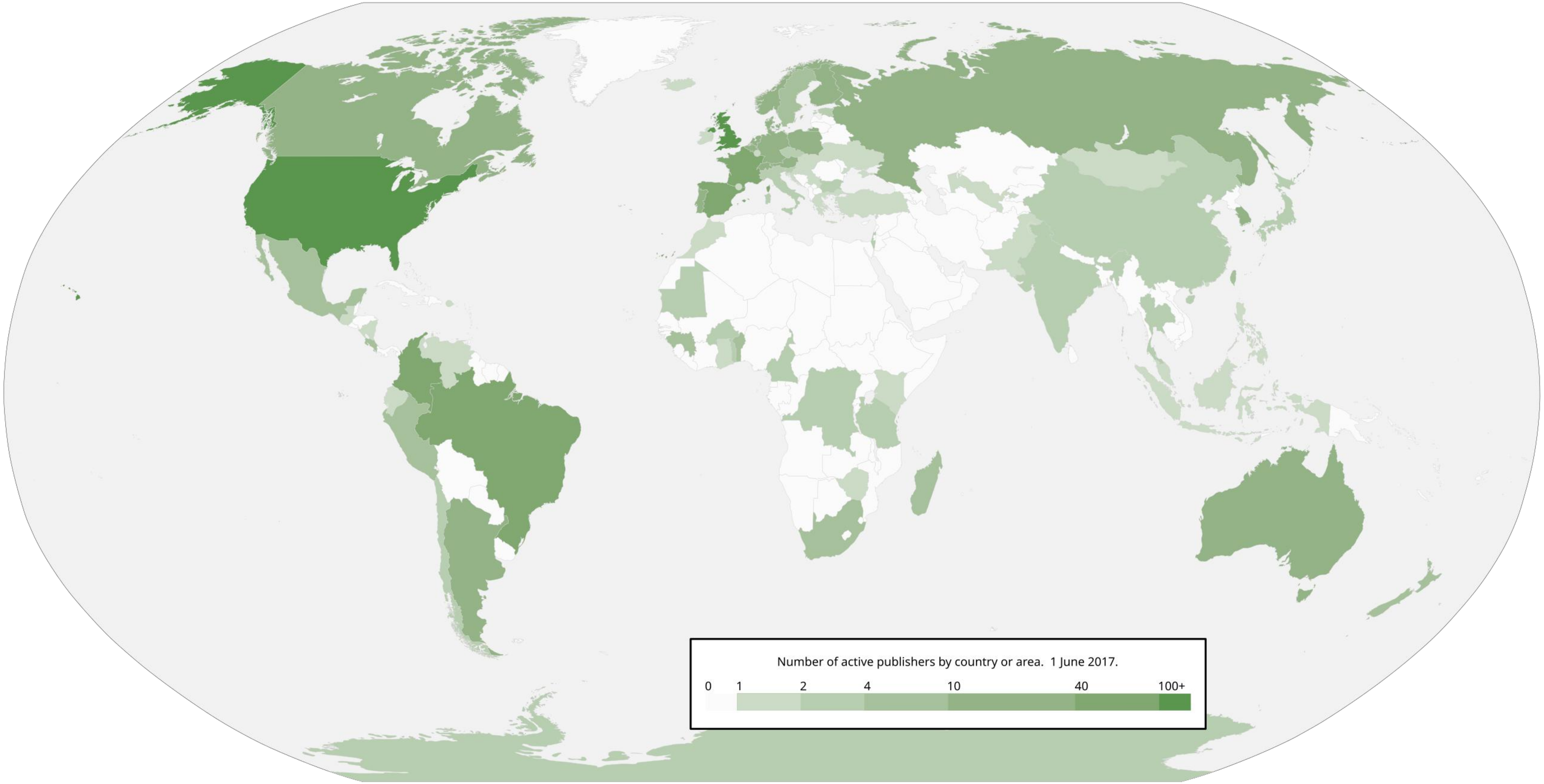
Projects funded through BID, CESP and BIFA 2015-2025



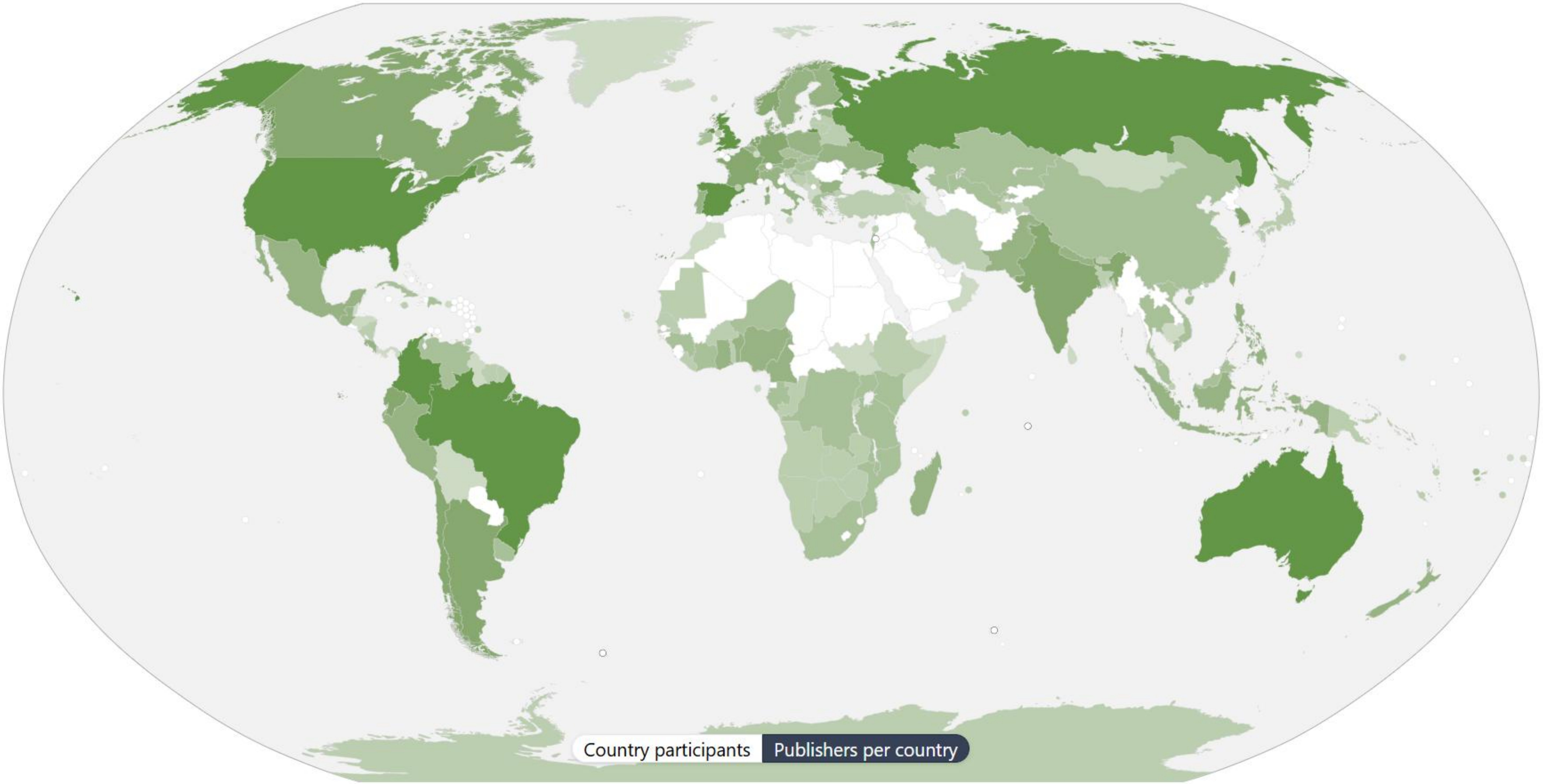
The BID programme is funded by the European Union

Funded by the European Union

Number of active data publishing institutions 2017

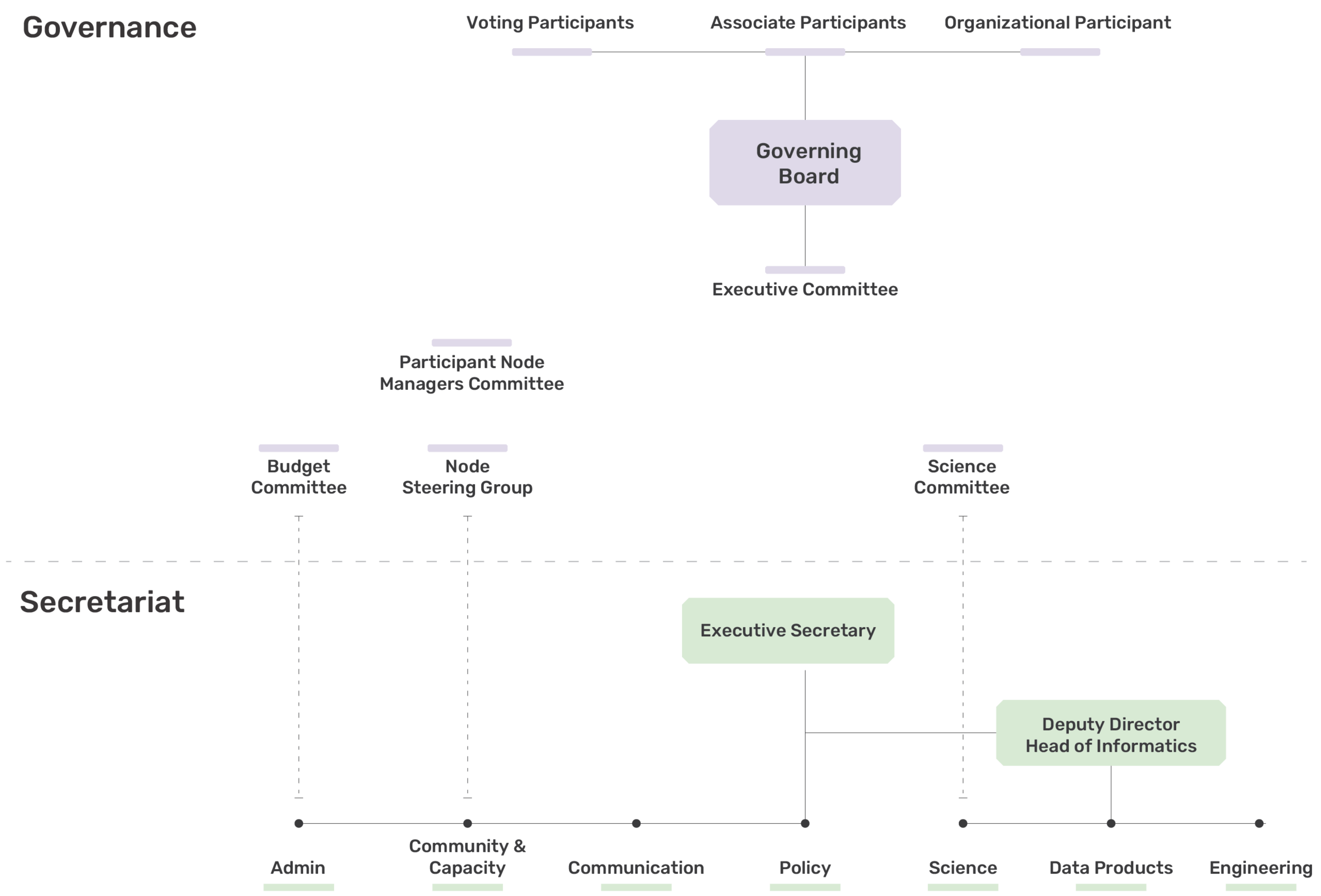


Number of active data publishing institutions 2026



Supporting and developing GBIF nodes through collaboration

GBIF’s governance model is an important strength



GBIF France has contributed to all GBIF’s governance structures and standing committees over its 20 years!

Anne-Sophie Archambeau currently 3rd Vice Chair of Governing Board

Acknowledging contributions to the nodes committee

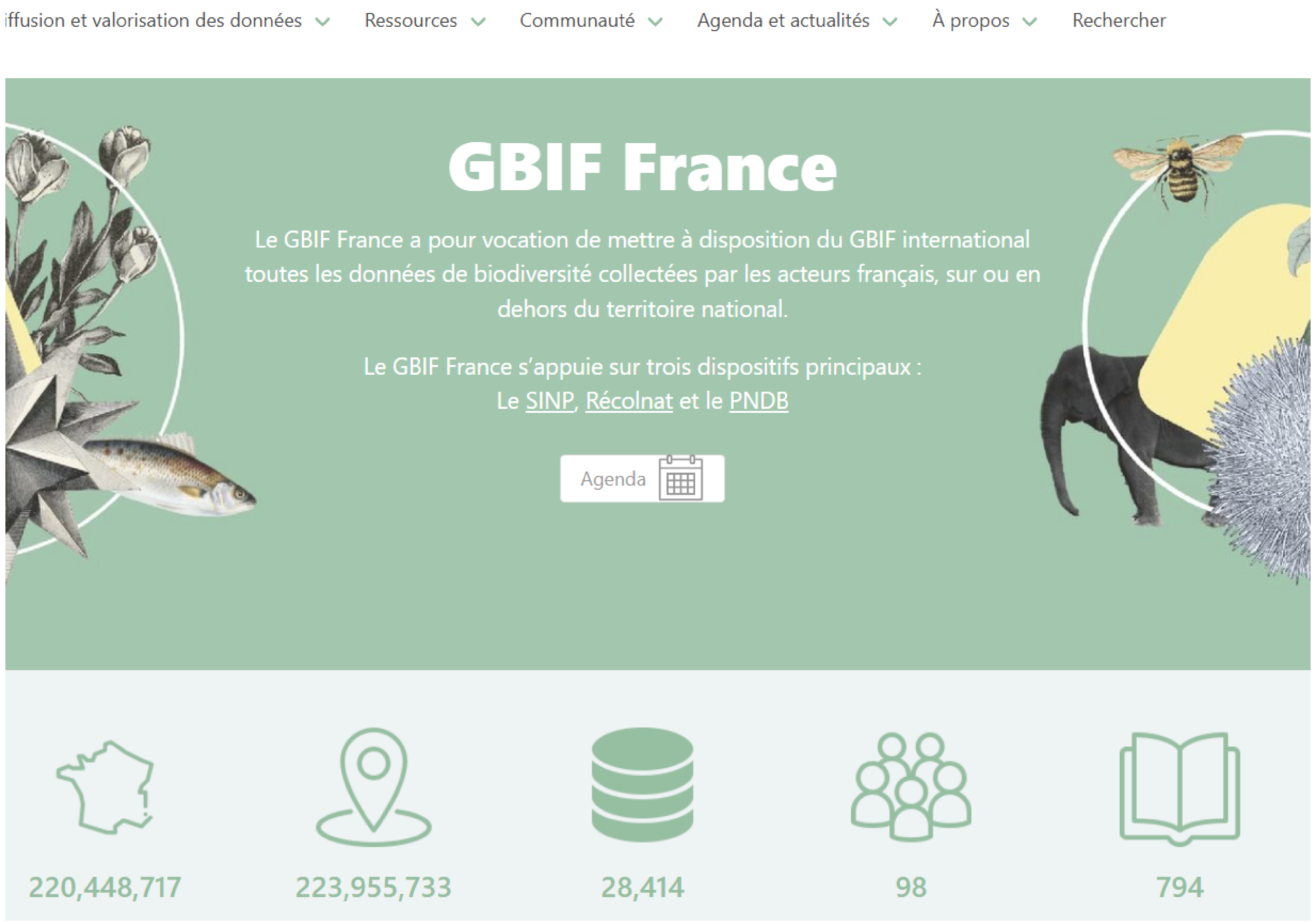
Anne-Sophie Archambeau chaired the nodes committee for a record 5 years

GBIF France has contributed to many mentoring actions

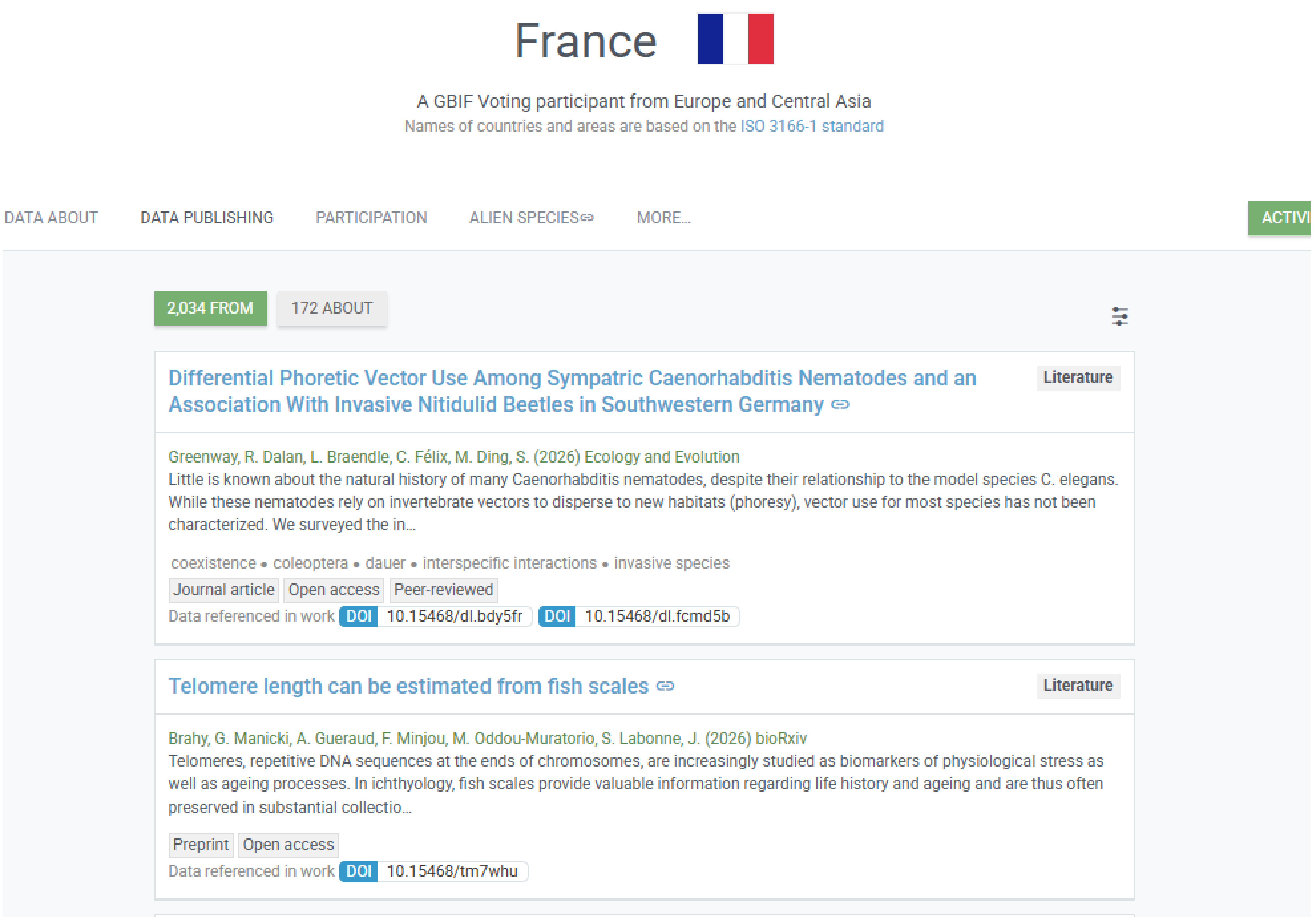
GBIF France active at the regional level, hosting the latest regional meeting in Montpellier, May 2026

Hosted infrastructure services

- Lowering technical threshold
- GBIF France inspired us to provide hosted IPTs (data publishing tools)
- GBIF France now runs a hosted portal
- Hosted portals for scientific journals of MNHN



Stronger together – benefits of participating in a global network



France

A GBIF Voting participant from Europe and Central Asia
Names of countries and areas are based on the ISO 3166-1 standard

DATA ABOUT | DATA PUBLISHING | PARTICIPATION | ALIEN SPECIES | MORE... | ACTIVI

2,034 FROM | 172 ABOUT

Differential Phoretic Vector Use Among Sympatric Caenorhabditis Nematodes and an Association With Invasive Nitidulid Beetles in Southwestern Germany — Literature

Greenway, R. Dalan, L. Braendle, C. Félix, M. Ding, S. (2026) Ecology and Evolution
Little is known about the natural history of many Caenorhabditis nematodes, despite their relationship to the model species C. elegans. While these nematodes rely on invertebrate vectors to disperse to new habitats (phoresy), vector use for most species has not been characterized. We surveyed the in...

coexistence • coleoptera • dauer • interspecific interactions • invasive species
Journal article | Open access | Peer-reviewed
Data referenced in work DOI 10.15468/dl.bdy5fr DOI 10.15468/dl.fcmd5b

Telomere length can be estimated from fish scales — Literature

Brahy, G. Manicki, A. Gueraud, F. Minjou, M. Oddou-Muratorio, S. Labonne, J. (2026) bioRxiv
Telomeres, repetitive DNA sequences at the ends of chromosomes, are increasingly studied as biomarkers of physiological stress as well as ageing processes. In ichthyology, fish scales provide valuable information regarding life history and ageing and are thus often preserved in substantial collectio...

Preprint | Open access
Data referenced in work DOI 10.15468/tm7whu

- Visibility
- Knowledge sharing in expert network
- Collaborative projects
- Data repatriation
- Supporting national research and reporting

www.gbif.org/country/FR/publications/from

As GBIF's relevance expands...

Thematic communities

Data model work

Relevance to policy frameworks

... so does the need for strong nodes

Coordinating expanding networks

New communities to train

Developing data flows

Merci!

mraymond@gbif.org